

1. Describe CCNOT (Toffoli) gate with its application.

The CCNOT (Controlled-Controlled) gate commonly known as the Toffoli gate is a fundamental building block in both quantum computing and reversible computing systems. It was introduced by physicist Tommaso Toffoli and is considered one of the most important multi-qubit gates. And the Toffoli gate is

- * Reversible
- * Information preserving
- * Energy efficient.

It operates on three qubits:

- Two control qubits
- One target qubit

Basic concept and working:

Let the inputs be:

- First control qubit → A
- Second control qubit → B
- Target qubit → C

Operation rule:

- * If both A and B are 1, then flip C.

* Otherwise, leave C unchanged.

Logical expression:

$$C' = C \oplus (A \cdot B)$$

$$A' = A, B' = B$$

Truth table Explanation:

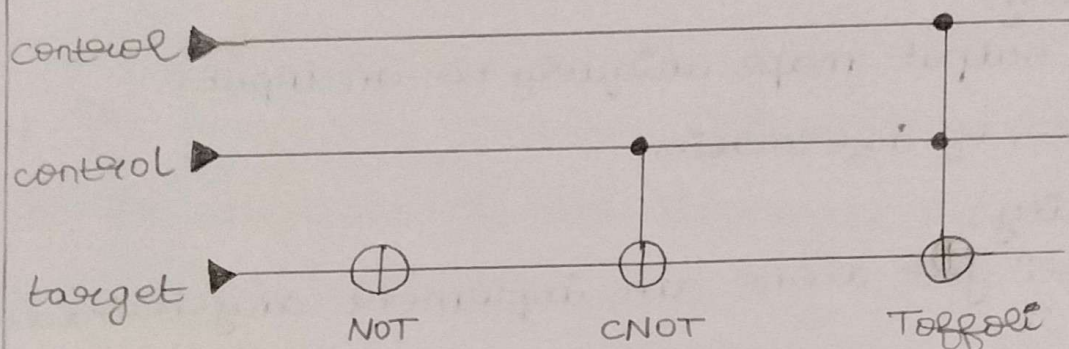
A	B	C	A'	B'	C'
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	1
1	1	1	1	1	0

→ For all rows except when $A=1$ and $B=1 \rightarrow C$ remains same.

→ When $A=1$ and $B=1 \rightarrow C$ is flipped

⇒ Hence, the gate behaves like a controlled-controlled switch.

Circuit Representation:



Interpretation:

- control qubits send signals vertically
- Target flips only when both controls are active.

Quantum Mechanical Representation:

The Toffoli gate operates on quantum states:

$$|A, B, C\rangle \rightarrow |A, B, C \oplus (A, B)\rangle$$

Basic State Transformation:

$$* |110\rangle \rightarrow |111\rangle$$

$$* |111\rangle \rightarrow |110\rangle$$

All other states remain unchanged.

Matrix Representation:

Since it operates on 3 qubits, the matrix is 8×8 . It is an identity matrix except for swapping:

$$* |110\rangle \text{ and } |111\rangle$$

This ensures:

→ Unitary property

→ Reversibility

* Every output maps uniquely to an input.

* No loss of information.

Important properties:

→ Reversibility:

- * Every output maps uniquely to an input.
- * No loss of information

→ Universality:

- * Toffoli gate alone can implement any Boolean function.

* Hence called a universal reversible gate.

→ Energy efficiency:

- * Avoids heat loss due to bit erasure.
- * Important in low-power computing.

→ Deterministic behavior:

- * Works predictably based on input conditions.

Application of Toffoli Gate:

* Reversible Computing

* Quantum computing

→ Quantum Arithmetic Circuits

→ Quantum Algorithms

→ Quantum control logic

* Low power VLSI Design

* Cryptography

* Digital Circuit Design

Advantages:

* No information loss

* Energy efficient

* Universal Gate

* Supports Quantum Computation

* Scalable for complex systems.

2. Explain Quantum Fourier Transform (QFT) and its application.

The Quantum Fourier Transform (QFT) is the quantum analogue of the Discrete Fourier Transform (DFT) used in classical computing. It is one of the most important operations in quantum computing, forming the core of several powerful quantum algorithms.

The QFT transforms a quantum state from computational basis into the frequency domain, enabling efficient extraction of periodicity and phase information.

Basic concept of QFT:

Consider an n -qubit quantum system which can represent 2^n states:

$$|\alpha\rangle \text{ where } \alpha = 0, 1, 2, \dots, 2^n - 1$$

The QFT maps this state into a superposition of all basis states:

$$|\alpha\rangle \rightarrow \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{\frac{2\pi i \alpha k}{2^n}} |k\rangle$$

This transformation introduces phase factors that encode frequency information.

Mathematical Representation:

For an input quantum state:

$$|\psi\rangle = \sum_{x=0}^{N-1} a_x |x\rangle$$

After applying QFT:

$$\text{QFT}(|\psi\rangle) = \sum_{k=0}^{N-1} \left(\frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} a_x e^{2\pi i x k / N} \right) |k\rangle$$

where

- * $N = 2^n$

- * a_x are amplitudes.

Working Principle of QFT:

The QFT works by applying a sequence of

1. Hadamard Gates (H)

- * Create superposition

- * Introduce phase relationships

2. Controlled Phase Rotation Gates

- * Add phase shift based on qubit positions.

- * Capture frequency information.

3. Swap Operations

- * Reverse the order of qubits.

Complexity of QFT:

- * Classical DFT complexity: $O(N^2)$

- * Fast Fourier Transform (FFT): $O(N \log N)$

- * QFT complexity: $O(n^2)$

Since $N = 2^n$, QFT is exponentially faster than

classical DFT.

Properties of QFT:

→ Unitary operation:

- * QFT is reversible

- * preserves quantum information

→ Linearity:

- * works on superposition of states

→ Efficient Implementation:

- * requires fewer gates than classical methods

→ Phase encoding

- * Encodes information in phase rather than amplitudes.

Applications of QFT:

- * Shor's Algorithm

- * Phase Estimation Algorithm

- * Quantum Signal Processing

- * Quantum Algorithms.

- * Quantum Cryptography.

- * Solving Linear systems.

- * Quantum Simulation

Conclusion:

The QFT is a cornerstone of quantum computing. Its ability to efficiently transform quantum states into the frequency domain. It plays a vital role in

- * Shor's algorithm

- * Phase estimation

- * Quantum Simulations.

3. Explain Deutsch-Jozsa algorithm and compare it with classical approach.

The Deutsch-Jozsa algorithm is one of the earliest and most important examples of a quantum algorithm that demonstrates a clear advantage over classical computation.

Problem Statement:

Given a function:

$$f: \{0, 1\}^n \rightarrow \{0, 1\}$$

Determine whether the function is:

- * Constant $\rightarrow$ same output (all 0s or all 1s), OR
- * Balanced $\rightarrow$ outputs 0 for exactly half the inputs and 1 for other half.

Classical Approach:

Working:

In classical computation:

* You must evaluate the function $f(x)$ for different inputs.

* Compare outputs to determine whether it is constant or balanced.

Worst case:

$\rightarrow$ Need to check more than half of the inputs

That is:

$$2^{n-1} + 1 \text{ evaluations.}$$

Limitations:

→ Exponential complexity

$$\text{Time complexity} = O(2^n)$$

→ Inefficient for large n

Number of evaluations grow rapidly.

→ No Deterministic Shortcut

Cannot decide in fewer steps in worst case.

Quantum Approach: Deutsch-Jozsa Algorithm:

The Deutsch-Jozsa algorithm uses quantum parallelism and interference to solve the problem efficiently.

Key Advantage:

→ Requires only one oracle query

→ Deterministic result.

Steps of Deutsch-Jozsa Algorithm:

→ Initialization

Prepare n input qubits initialized to

$$|0\rangle^{\otimes n}$$

Prepare 1 auxiliary qubit

$$|1\rangle$$

Initial state:

$$|0\rangle^{\otimes n} |1\rangle$$

→ Apply Hadamard Gates

Apply Hadamard (H) to all qubits

Input qubits become:

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

Auxiliary qubit becomes:

$$\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Resulting state:

$$\frac{1}{\sqrt{2^n}} \sum_x |x\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

→ Oracle function (U_f)

The oracle implements:

$$U_f |x, y\rangle = |x, y \oplus f(x)\rangle$$

Effect:

→ Introduces a phase shift:

$$\text{New state} \quad (-1)^{f(x)}$$

$$\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle$$

→ Apply Hadamard Again

This step causes:

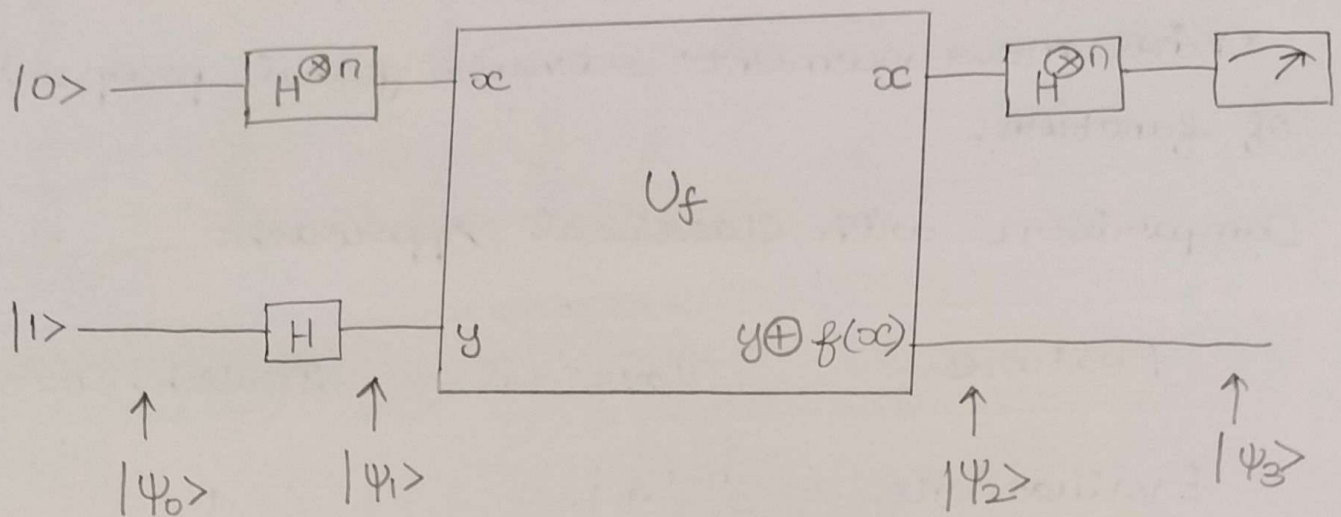
- * Constructive interference (for constant functions)
- * Destructive interference (for balanced functions)

→ Measurement

Measure the input qubits

- * If output = $|0 \dots 0\rangle \rightarrow$ function is constant
- * Otherwise $\rightarrow$ function is balanced.

Circuit Representation:



Example:

$n=2$

1: Constant function

$f(x)=0$ for all x

Result:

Final output $\rightarrow |00\rangle$

Detected as constant

2: Balanced function

$f(00)=0, f(01)=1, f(10)=1, f(11)=0$

Result:

Output $\neq |00\rangle$

Detected as balanced.

Why the Algorithm works:

The algorithm relies on:

1. Quantum Parallelism:

$\rightarrow$ Evaluates all inputs simultaneously

2. Phase Encoding:

$\rightarrow$ Stores function output in phase

3. Interference

→ Cancels or reinforces amplitude.

* Final measurement reveals global property of function.

Comparison with classical Approach:

Feature	Classical	Deutsch-Jozsa
Evaluations	$2^{n-1} + 1$	1
Time complexity	$O(2^n)$	$O(n)$
Nature	Deterministic	Deterministic
Efficiency	Poor for large n	Highly efficient
Parallelism	No	Yes
Scalability	Limited	Excellent.

Applications:

→ Quantum Algorithm Design

→ Complexity Theory

→ Quantum Circuit Optimisation

→ Education and Research.